Safety systems · Netherlands

Dutch signals show the **speed**, not the route: green, yellow or red, with a lit **number × 10 km/h** below. **Flashing green** = the speed applies **from this signal**; **yellow + number** = **reach it by the next signal**; a flashing number or flashing yellow are special cases (see the cards). In TSW 7: **NS ICMm** and **NS SNG** on **Zwolle – Groningen**, under **ATB-EG**, **ATB-Vv** and the **deadman**. Quadrant controls are from TheJAG's profiles.

Safety systems

ATB-EG
Automatische Treinbeïnvloeding, eerste generatie

Coded track circuits set a ceiling of 40, 60, 80, 130 or 140 km/h. Overspeed rings a bell; brake within ~**3 s** or ATB brakes.

WHERE Whole Zwolle – Groningen line; ICMm and SNG

QUADRANT Left aux square: ATB fuse (ICMm) / ATB / ERTMS cut-out (SNG)

ATB-Vv
ATB verbeterde versie

Beacons 120, 30 and 3 m before selected red signals impose a braking curve to a stop. Acts only under code [–] or [BD].

WHERE Selected stop signals on Zwolle – Groningen

QUADRANT Shares the ATB fuse / cut-out (left aux square)

Deadman
Dodemansinrichting

Driver-alive check. Answer the warning in time or the train makes an emergency brake application.

WHERE ICMm and SNG

QUADRANT Gear lever up = acknowledge (game key Q); right aux square = deadman fuse / cut-out

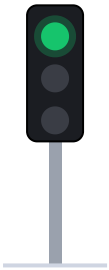
ATB-NG · ETCS
ATB nieuwe generatie · European Train Control System

ATB-NG (balises, 10 km/h steps) on Dutch diesel lines; ETCS on Betuweroute, HSL-Zuid, Hanzelijn. Not in TSW 7.

WHERE Not on any TSW Dutch route

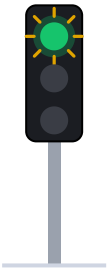
Main signal aspects (hoofdseinen)

Lamps top to bottom: green, yellow, red; white number below. Dwarf signals: same colours, max 40 km/h.



Green (groen)
Proceed at the local (line) speed. ATB code [13] or [14].

DRIVE Accelerate to the speed on the boards.



Flashing green
Proceed at max 40 km/h from this signal. ATB code [–] 40.

DRIVE Be at 40 km/h passing the signal; usually a diverging route.



Flashing green + 8
Proceed at max 80 km/h from this signal (number × 10).

DRIVE Hold 80 km/h until a speed board lets you go faster.



Yellow (geel)
Next signal may be red. Limit speed to 40 km/h. ATB code [–] 40.

DRIVE Brake now to 40 km/h and be ready to stop at the next signal.



Yellow + 8
Reduce to 80 km/h, reached before the next signal. ATB code [8].

DRIVE Brake to 80 km/h before the next signal.



Yellow + flashing 6
Limit to 60 km/h; short section, may not be reachable by the next signal.

DRIVE Brake to 60; keep braking if the next signal asks for less.



Flashing yellow
Drive on sight (rijden op zicht): track beyond may be occupied. Max 40 km/h.

DRIVE Creep, max 40 km/h, able to stop short of any obstruction.



Red (rood)
Stop before the signal. ATB-EG does not stop you; ATB-Vv may.

DRIVE Stop short of the signal; wait for a proceed aspect.

Passing red: ATB-EG never enforces below 40 km/h, so it lets you pass a red signal below about 45 km/h (40 + 5). Stop short yourself; in TSW only ATB-Vv beacons can save you, and only at selected signals.

Sources: Regeling spoorverkeer, Bijlage 4 (Seinen), wetten.overheid.nl · Dovetail Games: Zwolle – Groningen ATB guide · en.wikipedia: Dutch railway signalling · en.wikipedia: Automatische Treinbeïnvloeding · nl.wikipedia: Automatische treinbeïnvloeding · OV in Nederland Wiki: Seinen · IRSE Nederland: Seinen · Dovetail Games forums

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Safety systems · Netherlands

Boards and other signals

Board numbers are × 10 km/h.



P-sein

Automatic block signal (P-bord). Red still means stop.

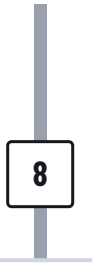
DRIVE Call the signaller; pass on sight only if no contact is possible.



Speed reduction

Snelheidsverminderingsbord: lower limit ahead.

DRIVE Be at 80 km/h by the next white speed board.



Speed board

Snelheidsbord: local speed beyond the board.

DRIVE Lower: from the front. Higher: once the rear has passed.



Line speed

Green triangle (sein 316): line speed may be resumed.

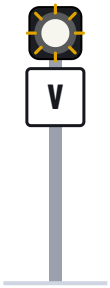
DRIVE Accelerate once the rear of the train has passed.



Train length

Treinlengtebord: stop marker for trains of up to this many cars.

DRIVE Stop at the board for your vehicle count, or the next one up.



Departure light

Vertrekseinlicht: the departure procedure may start.

DRIVE Wait for it (and a proceed aspect) before the doors close.

ATB-EG cab codes

Code	Ceiling (+5 tolerance)	Usually sent by	Cab display
[14]	140 km/h	Green on a 140 line	Green lamp / green mark
[13]	130 km/h	Green, or a number 9–13	Yellow 13
[8]	80 km/h	Number 7 or 8 (yellow or flashing green)	Yellow 8
[6]	60 km/h	Number 5 or 6	Yellow 6
[–]	40 km/h	Yellow, flashing green, flashing yellow, red	Lowest yellow lamp
[BD]	No limit	ATB isolated or failed (buiten dienst)	Blue lamp, five bells

How to drive under ATB-EG and the deadman

Code drops (speed reduction)

1. A single chime and the cab display shows the lower code, after a short random delay in TSW.
2. Already below the new ceiling: nothing to do.
3. Above it (+5 km/h): a continuous bell. Brake within about 3 s.
4. Make at least a minimum brake application (ICMm step 1 or more, SNG min brake or more); the bell stops.
5. Three short bells below ceiling +5: you may release the brake.

Brake on time for the signals anyway: ATB only checks you brake, not that you stop in time.

ATB brake (forced brake)

1. No brake within ~3 s, or overspeed: ATB brakes to a stop; red ATB lamp lit.
2. When stopped, set full brake on the right throttle: ICMm step 7, SNG Max brake.
3. Press Ontgrendelen (unlock) in the cab with the mouse; it is not on the Quadrant.
4. Wait for the red lamp to go out, then release and drive on.

ATB-Vv at red signals

- Acts only under code [–] or [BD]; the beacons are not shown to you.
- After the 30 m beacon the limit is 10 km/h (+5).
- Over the braking curve: forced brake to a stop.
- A release loop up to 60 m before the signal, or a new code above [–], lifts the curve.

Deadman (dodemansinrichting)

1. Keep the deadman fuse on: right aux square (ICMm fuse, SNG cut-out in).
2. When the warning sounds, lift the gear lever (game key Q).
3. Miss it and the train brakes; acknowledge, then release and drive on (TSW behaviour not yet verified).

The real system is a held pedal or a periodic button; TSW reacts to Q.

Before departure

1. Left aux square on: ATB fuse (ICMm) or ATB / ERTMS cut-out set to cut in (SNG). Off gives [BD] and five bells.
2. Right aux square on: deadman fuse / cut-in.
3. Check the cab code matches the signal, then pull away.

Boards to know

- Verkenbord: platform ahead at braking distance; start braking for the stop.
- CODE board: ATB-EG / ATB-NG changeover at the route fringes (ATB-NG itself is not simulated).
- A / E boards: start and end of a temporary speed restriction.

Sources: Regeling spoorverkeer, Bijlage 4 (Seinen), wetten.overheid.nl · Dovetail Games: Zwolle – Groningen ATB guide · en.wikipedia: Dutch railway signalling · en.wikipedia: Automatische Treinbeïnvloeding · nl.wikipedia: Automatische treinbeïnvloeding · OV in Nederland Wiki: Seinen · IRSE Nederland: Seinen · Dovetail Games forums

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